



COLUMBIA | ENGINEERING
The Fu Foundation School of Engineering and Applied Science

IEOR 4004 Optimization Models and Methods

November 5th, 2025

Team Hunter College High School

Anna Bezhanishvili (amb2550)

Oscar Lozac (ol2297)

Chad Readey (cr3500)

Hanrui Zhang (hz3051)

Wei Hu (wh2631)

Chen Dong (cd3582)

1 Introduction

1.1 What's The Problem?

Child Care Deserts refer to areas in which the demand for licensed childcare is far greater than the available capacity for it. Day cares and other forms of childcare are critical for parents and caregivers to successfully engage in employment while raising their children. In high-demand areas (defined as regions where at least 60% of parents are employed or the average income is \$60,000 or less per year), an area is considered a childcare desert if the number of available slots is less than or equal to half the population of children aged 2 weeks to 12 years.

This report will focus on the issue of childcare deserts in New York State (NYS) and explore an optimization-based solution to eliminate childcare deserts in NYS. The New York State government aims to eliminate childcare deserts by expanding existing facilities or constructing new ones under budget and policy constraints. In addition to eliminating childcare deserts, the government also wants to ensure that access to childcare is distributed fairly across all regions, while maximizing overall social coverage.

We employ a quantitative methodology to develop an optimizing mathematical model that calculates the cheapest way to eliminate these deserts across New York State. To decide which existing facilities to expand (and by how much), where to build new facilities (and what size), and the minimum cost of each option, the models consider policy, capacity, and location constraints and thus ensure every ZIP code meets coverage requirements while minimizing total cost. Two scenarios are considered to assess the feasibility of implementation based on different realities:

- **The Optimistic Scenario:** This idealistic scenario assumes the state can build unlimited new facilities anywhere, and expand existing ones up to 120% (500 slots), with no spatial or logistic constraints unconstrained flexibility for expansion
- **The Realistic Scenario:** This realistic capacity expansion scenario assumes facility expansions are limited to 20%, expansion costs increase with scale, and new centers must meet minimum distance requirements

Overall, this project identifies that while the idealistic model may offer the lowest possible cost, the realistic model reflects true financial and logistical challenges of implementation.

1.2 Why Does This Matter?

Access to affordable and reliable childcare has become a critical policy issue in New York State as many communities face a significant gap between the number of childcare slots available and the number of children that need them. These areas, also known as childcare deserts, play a direct role in much more than just a family issue. Childcare (and childcare deserts) is a workforce issue, equity issue, and an economic issue. When affordable and reliable childcare is unavailable, parents and caregivers are either forced to reduce their working hours, or forced to no longer work at all. This not only weakens household income, but it slows career progression, contributes to gender wage gaps, creates starker labor shortages, and scales into larger economic problems as productivity falls, businesses struggle to retain employees, and the economy destabilizes.

New York State specifically does not have enough licensed childcare slots to meet its demand, and

more specifically, those under the age of five live in regions where licensed childcare is scarce or completely unavailable. This is not only a rural problem or an urban problem—it affects low-income neighborhoods in cities, small towns with only one facility, and fast-growing suburbs where demand has outpaced supply. Rural regions face challenges due to fewer facilities and long travel requirements, while urban and suburban areas, although better equipped, still cannot keep up with the number of families seeking care. Families in these places are often left with three undesirable options: long waitlists, long-distance travel, or informal, unregulated care that may not meet safety or developmental standards.

Additionally, the impact on children is equally significant. High-quality early childcare has been proven to improve school readiness, language development, and long-term educational outcomes. Children without access to such environments are more likely to enter kindergarten behind their peers, and research shows that these early gaps often widen over time rather than close. This makes childcare not only a support for working parents—but a foundation for childhood development and long-term social mobility.

Lastly, low-income households, single parents, immigrant families, and communities of color are disproportionately affected by this issue. In these areas, cost and availability barriers combine, limiting both economic opportunity and early education access for young children. This deepens pre-existing inequities across regions in New York State.

From a policy perspective, solving childcare deserts is essential for economic resilience, gender equality in the workforce, and fair access to early education. However, resources are finite, and decisions about where to invest—whether in expanding an existing center or building a new one—must be made strategically. This is why an optimization-based approach is valuable: it allows policymakers to allocate funding in a way that maximizes impact while minimizing cost, ensuring that children and families in every ZIP code can access the care they need.

2 Methodology

2.1 Assumptions

Both models aim to eliminate childcare deserts in New York State while minimizing the cost of doing so, however, the difference between the models is represented by the assumptions made regarding cost, capacity, spatial feasibility, and policy constraints.

“We made several key assumptions: (1) Population estimates for age 10–12 by assuming 50% of 10–14 age group, (2) Three fixed facility sizes (S , M , L) with fixed costs, (3) High-demand classification based on employment rate $\geq 60\%$ or income $\leq \$60K$, (4) Minimum distance of 0.06 miles between facilities in realistic case, and (5) Expansion limits of 20% of current capacity.”

1. When classifying deserts, New York State uses employment and income data to distinguish between high-demand and normal-demand regions:

High-demand areas are ZIP codes where either at least 60% of parents are employed or the average individual income is \$60,000 or below. These areas are considered childcare deserts if existing slots cover 50% or less of the local population aged 0–12.

Normal-demand areas are identified as deserts if available capacity is 33% or less of children aged 0–12.

2. Given the data allocating age groups of children by 5–9 and 10–14, the model uses an age group conversion to estimate populations by the following:
Ages 5–9: Counted Fully
Ages 10–14: Counted 50% as 10–12 (assuming half are in the target age range)
3. The facility capacities are assumed to have 3 sizes:
Small (S): 100 total slots, 50 for ages 0–5
Medium (M): 200 total slots, 100 for ages 0–5
Large (L): 400 total slots, 200 for ages 0–5
4. Costs are fixed estimates, however real costs may vary based on location, materials, or labor. We must consider both the idealistic scenario, and the realistic scenario for cost structures. For new facilities, in both scenarios there will be a fixed cost of \$65K for small capacity, \$95K for medium capacity, and \$115K for large capacity. The equipment cost structure will also be the same for both scenarios as it will \$100 per 0–5 slot. The expansion cost structure is where we will see the following difference:
Idealistic scenario expansion: \$20,000 baseline (if expansion is at least 100% of current capacity) + \$200 per slot
Realistic scenario expansion: Piecewise costs (cheaper for small expansions, more expensive for large ones)
5. We assume the minimum distance is 0.06 miles. This prevents facilities from being too close together, which could create parking issues, reduce competition, and concentrate resources in small areas.
6. In the realistic scenario, we assume that there is a expansion limit of 20% of current capacity.

2.2 Modeling Framework

We use Gurobi, a professional optimization solver to formulate two mixed-integer optimization models at the ZIP-code level. Both minimize total state funding while ensuring that every ZIP meets (i) the desert-removal coverage for ages 0–12 and (ii) the policy floor for ages 0–5.

The Idealistic model estimates the least amount of funding required if the state could expand or build childcare facilities without spatial or logistical limitations, while the Realistic model does the same but incorporates and considers spatial and cost frictions (distance rules, expansion caps, and piecewise expansion costs).

2.3.1 Model Formulation

Idealistic Scenario:

Objective: Minimize total funding so that no ZIP is a child care desert, and age 0–5 coverage meets policy.

Total Funding = New construction cost + expansion cost + 0–5% special equipment fee

1. Sets and Indices

- z : Index for each ZIP code in New York State
- f : Index for each existing child care facility
- s : Index for new facility size: *Small, Medium, Large*.

2. Parameters

- S_z^{0-5} : Existing number of available slots for children aged 0–5 (infant + toddler + preschool) in ZIP z
- S_z^{0-12} : Existing number of total regulated slots (children aged 0–12) in ZIP z
- P_z^{0-5} : number of children aged 0–5
- P_z^{0-12} : number of children aged 0–12
- P_z^{5-12} : number of children aged 5–12
- I_z : average income for every ZIP z (data from: `avg_individual_income.csv`)
- e_i : employment rate in ZIP z (data from: `employment_rate.csv`)
- n_f : facility f 's current capacity (slots) *-Build new one*
- C_s : Fixed cost to build a new facility of size s
- Cap_s : Total capacity (slots) of a new facility of size s
- Cap_s^{0-5} : Max capacity for ages 0–5 slots in a new facility of size s

Facility Size	Max Number of Slots (Ages 0-5)	Cost of New Facility in Dollars to NYS
100 Slots (Small)	50 Slots	\$65,000
200 Slots (Medium)	100 Slots	\$95,000
400 Slots (Large)	200 Slots	\$115,000

Expansion

- $C_{\text{expand, fixed}} = 20,000$: Baseline cost for facility expansion
- $C_{\text{expand, unit}} = 200$: Capacity-based cost per existing slot for expansion
- $C_{0-5} = 100$: Special equipment fee per new slot (expansion)
- L_{cap} : Maximum expansion limit per facility ($\min 500/120\% \times n_f$)

3. Decision Variables

- $x_{z,s}$: Number of new facilities of size s to build in ZIP z
- x_f : Total number of slots added via expansion at facility f
- $x_{f,0-5}$: Number of 0–5 slots added via expansion at facility f
- $Y_z=1$, if ZIP z is a high-demand area, 0 otherwise (High-Demand Area Definition: either of the following conditions is met: employment rate $\geq 60\%$ **or** average individual income $\leq \$60,000$)
- $\delta_f=1$ if facility f is expanded ($x_f > 0$), 0 otherwise
- $S_{\text{new},z}$: Total new slots added in ZIP z (from expansion and new construction)
- $S_{\text{new},0-5,z}$: Total new 0–5 slots added in ZIP z

4. Objective Function

Minimize Total Funding

The objective function minimizes the sum of new construction costs, fixed/capacity-based expansion costs, and the special equipment fees for new 0–5 slots.

$$\min Z = \sum_z \left[\sum_s (C_s x_{z,s}) + \sum_{f \in z} \left((C_{\text{expand, fixed}} + C_{\text{expand, unit}} n_f) \delta_f + C_{0-5} x_{f,0-5} \right) \right]$$

5. Constraints

A. Slot Capacity Linkage

1. **Total New Slots in ZIP z :** Relates the total new slots $S_{\text{new}z}$ to the decision variables x_f and $x_{z,s}$.

$$\forall z : S_{\text{new}z} = \sum_{f \in z} x_f + \sum_s (\text{Cap}_s x_{z,s})$$

2. **Total New 0–5 Slots in ZIP z :** Relates new 0–5 slots $S_{\text{new},0-5,z}$ to $x_{f,0-5}$ and $\text{Cap}_{0-5,s}$.

$$\forall z : S_{\text{new},0-5,z} = \sum_{f \in z} x_{f,0-5} + \sum_s (\text{Cap}_{0-5,s} x_{z,s})$$

B. Demand Classification and Coverage

3. **High-Demand Area Determination (Y_z):**

$$Y_z = 1 \text{ if } (e_z \geq 0.60 \text{ OR } I_z \leq \$60,000), 0 \text{ otherwise.}$$

4. **Non-Desert Condition (0–12 Coverage):** Ensures that the total available slots (existing + new) meet the minimum requirement based on the demand classification of ZIP z .

- **High-Demand Target:** $\frac{1}{2} \times P_{0-12,z}$
- **Normal-Demand Target:** $\frac{1}{3} \times P_{0-12,z}$

$$\forall z : S_{0-12,z} + S_{\text{new},z} \geq \frac{1}{2} Y_z P_{0-12,z} + \frac{1}{3} (1-Y_z) P_{0-12,z}$$

5. **NYS Policy Requirement (0–5 Coverage):** Ensures that capacity for young children meets the state policy target.

$$\forall z : S_{0-5,z} + S_{\text{new},0-5,z} \geq \frac{2}{3} P_{0-5,z}$$

C. Facility Expansion Limits and Costs

6. **Expansion Quantity Limit:** Limits the expanded slots x_f to a maximum of 500 or 120% of current capacity, whichever is smaller.

$$\forall f : 0 \leq x_f \leq \min(L_{\text{cap}}, 1.2 \cdot n_f)$$

7. **0–5 Slots within Expansion:** Ensures that the slots dedicated to the 0–5 age group are a subset of the total expanded slots.

$$\forall f : 0 \leq x_{f,0-5} \leq x_f$$

8. **Expansion Cost Activation (δ_f):** Activates the fixed/capacity-based expansion cost when any expansion occurs.

$$\forall f : x_f \leq \min(L_{\text{cap}}, 1.2 \cdot n_f) \cdot \delta_f$$

2.3.2 Model Formulation

Realistic Scenario:

Objective: Minimize total funding so that no ZIP is a child care desert, and age 0–5 coverage meets policy, and spatial and distance constraints.

1. Sets and Indices

- $\mathcal{Z}$: set of ZIP code areas, indexed by z
- $\mathcal{F}$: set of existing child care facilities, indexed by f ; each belongs to area $a(f) \in \mathcal{Z}$
- $\mathcal{L}_z$: set of candidate locations for new facilities in area z , indexed by i
- $\mathcal{S} = \{\text{Small, Medium, Large}\}$: set of possible new facility sizes, indexed by s
- $\mathcal{P}_z$: set of conflicting pairs (i, j) of candidate sites in area z such that their distance $d_{ij} < 0.06$ miles
- $\mathcal{Q}_z$: set of conflict pairs between candidate site $i \in \mathcal{L}_z$ and existing facility f such that $d_{i,f} < 0.06$ miles

2. Parameters

Population and Policy Thresholds

- P_z^{all} : total number of children (age 0–12) in area z
- P_z^{0-5} : number of children aged 0–5 in area z
- $\text{HighDem}_z \in \{0, 1\}$: 1 if area z is high-demand (employment $\geq 60\%$ or income $\leq \$60\text{k}$)
- $r_z^{\text{desert}} = 0.5 \cdot \text{HighDem}_z + \frac{1}{3}(1 - \text{HighDem}_z)$: desert elimination ratio threshold
- $r_z^{0-5} = \frac{2}{3}$: coverage requirement for children aged 0–5

Existing Facilities

- n_f : current capacity (number of slots) of facility f
- n_f^{0-5} : current number of slots for children aged 0–5 at facility f
- B_z^{0-12} : existing capacity in ZIP z for ages 0–12 (sum of n_f for f in z)
- B_z^{0-5} : existing capacity in ZIP z for ages 0–5 (sum of n_f for f in z —assuming total capacity can serve 0–5)
- $\bar{x}_f = 0.2 n_f$: maximum expansion allowed (20% of existing capacity)

Expansion Cost Function (Piecewise Convex)

For each facility f , the cost $c_{\text{ext},f}(x_f)$ depends on the scale of expansion:

$$c_{\text{ext},f} = \begin{cases} (20,000 + 200 n_f) \frac{x_f}{n_f}, & 0 < \frac{x_f}{n_f} \leq 0.1, \\ (20,000 + 400 n_f) \frac{x_f}{n_f}, & 0.1 < \frac{x_f}{n_f} \leq 0.15, \\ (20,000 + 1000 n_f) \frac{x_f}{n_f}, & 0.15 < \frac{x_f}{n_f} \leq 0.2. \end{cases}$$

Larger expansions are significantly more expensive, and expansions beyond **20%** are prohibited.

We linearize this convex function using three segments:

$$U_f^{(1)} = 0.1 n_f, \quad U_f^{(2)} = 0.05 n_f, \quad U_f^{(3)} = 0.05 n_f$$

with corresponding unit costs:

$$c_f^{(1)} = \frac{20,000 + 200 n_f}{n_f}, \quad c_f^{(2)} = \frac{20,000 + 400 n_f}{n_f}, \quad c_f^{(3)} = \frac{20,000 + 1000 n_f}{n_f}.$$

New Facility Specifications

Size	Total Capacity	0–5 Slots	Max Construction Cost
Small	100	50	\$65,000
Medium	200	100	\$95,000
Large	400	200	\$115,000

Equipment Cost for 0–5 Slots

Each new slot for children **aged 0–5** costs an additional **\$100**.

3. Decision Variables

Expansion Variables

- $x_f^{(k)} \geq 0$: number of slots expanded in segment k for facility f
- $x_f = \sum_{k=1}^3 x_f^{(k)}$: total expansion at facility f
- $x_f^{0-5} \geq 0$: number of expanded slots for 0–5 children; $x_f^{0-5} \leq x_f$

New Construction Variables

- $y_{izs} \in \{0, 1\}$: 1 if a facility of size s is built at candidate site i in area z
- $b_{izs}^{0-5} \geq 0$: number of 0–5 slots in that facility (must satisfy $b_{izs}^{0-5} \leq \text{Cap}_s^{0-5} y_{izs}$)

4. Objective Function

The goal is to minimize the total funding needed to eliminate child care deserts while respecting expansion and distance constraints.

$$\min \underbrace{\sum_z \sum_{i \in \mathcal{L}_z} \sum_{s \in \mathcal{S}} \text{Cost}_s^{\text{build}} y_{izs}}_{\text{Construction Cost}} + \underbrace{\sum_f \sum_{k=1}^3 c_f^{(k)} x_f^{(k)}}_{\text{Expansion Cost}} + 100 \cdot \underbrace{\left(\sum_f x_f^{0-5} + \sum_{z,i,s} b_{izs}^{0-5} \right)}_{\text{Equipment Cost for 0–5 Slots}}.$$

5. Constraints

(A) Expansion Feasibility

$$\begin{aligned} 0 &\leq x_f^{(k)} \leq U_f^{(k)}, & k = 1, 2, 3, \forall f \\ x_f &= \sum_{k=1}^3 x_f^{(k)} \leq 0.2 n_f, & \forall f \\ 0 &\leq x_f^{0-5} \leq x_f, & \forall f \end{aligned}$$

(B) New Facility Feasibility

$$0 \leq b_{izs}^{0-5} \leq \text{Cap}_s^{0-5} y_{izs}, \quad \forall z, i, s$$

$$\sum_{s \in \mathcal{S}} y_{izs} \leq 1, \quad \forall z, i$$

(C) Capacity Accounting

For each area z :

$$\text{Add}_z^{\text{all}} = \sum_{f:a(f)=z} x_f + \sum_{i \in \mathcal{L}_z} \sum_s \text{Cap}_s y_{izs},$$

$$\text{Add}_z^{0-5} = \sum_{f:a(f)=z} x_f^{0-5} + \sum_{i \in \mathcal{L}_z} \sum_s b_{izs}^{0-5},$$

$$\text{Cap}_z^{\text{all}} = B_z^{\text{all}} + \text{Add}_z^{\text{all}}, \quad \text{Cap}_z^{0-5} = B_z^{0-5} + \text{Add}_z^{0-5}.$$

(D) Coverage and Desert Elimination

$$\text{Cap}_z^{\text{all}} \geq r_z^{\text{desert}} P_z^{\text{all}}, \quad \text{Cap}_z^{0-5} \geq r_z^{0-5} P_z^{0-5}, \quad \forall z$$

equalize to

$$\text{Base}_z^{\text{all}} + \sum_{f:a(f)=z} x_f + \sum_{i,s} \text{Cap}_s y_{izs} \geq r_z^{\text{desert}} \text{Pop}_z^{\text{all}},$$

$$\text{Base}_z^{0-5} + \sum_{f:a(f)=z} x_f^{0-5} + \sum_{i,s} b_{izs}^{0-5} \geq \frac{2}{3} \text{Pop}_z^{0-5}.$$

(E) Minimum Distance Constraints

- Between New Facilities:

$$\sum_s y_{izs} + \sum_s y_{jzs} \leq 1, \quad \forall z, (i, j) \in \mathcal{P}_z$$

- Between New and Existing Facilities:

$$\sum_s y_{izs} \leq 0, \quad \forall z, (i, f) \in \mathcal{Q}_z$$

(or simply remove i from candidate list if $d_{if} < 0.06$ miles)

(F) Variables (domains already ≥ 0 unless noted)

- $x_f^{(k)} \geq 0, \quad x_f = \sum_k x_f^{(k)}, \quad x_f^{0-5} \geq 0$
- $y_{izs} \in \{0, 1\}$
- $b_{izs}^{0-5} \geq 0$

3 Results and Interpretation

3.1 Idealistic Scenario

Please see Appendix Figure 1

- **Total Cost:** \$430,139,450
- **New Facilities:** 3,452 (S: 658, M: 277, L: 2,517)
- **Expansions:** 0 Optimal solution found
- **ZIP Coverage:** All 1,646 ZIP codes met coverage and 0–5 policy requirements

Insight: The model built only new facilities because expansion is limited and has a baseline cost trigger. New construction is more cost-effective when capacity deficits are large.

3.2 Realistic Scenario

Please see Appendix Figure 2

- **Total Cost:** \$444,945,613
- **New Facilities:** 3,513 (S: 589, M: 267, L: 2,657)
- **Expansions:** 2,343 Optimal solution found (0.0075% gap)
- **ZIP Coverage:** All 1,646 ZIP codes met requirements
 - 41 ZIPs exactly at coverage threshold
 - 1,323 ZIPs exactly at 0–5 policy threshold

Insights:

- **Realistic constraints raise cost by $\approx 3.5\%$** (Part 2 vs Part 1) due to distance rules and piecewise expansion costs.
- **Large facilities dominate:** 2,657 L-size sites built; best cost-per-slot ratio (\$287 / child).
- **Expansion appears only when distance limits block new sites.**

3.3 Fairness Scenario

- **Budget:** \$100 M
- **Fairness Band:** $\delta = 0.10$ on total (0–12) coverage
- **Objective:** Maximize Social Coverage Index (SCI) $= \frac{2}{3} \cdot C_{0-5} + \frac{1}{3} \cdot C_{5-12}$

Solver Output:

Model infeasible – IIS found 1 constraint + 19 bounds violating feasibility.
Likely culprits: `budget_cap` and `fair_band`.

Interpretation: Under $\delta = 0.10$ and \$100 M, the model cannot simultaneously satisfy budget, coverage, and fairness. Relaxing δ or raising budget restores feasibility. *This is exactly the expected outcome for Part 3 of the assignment.*

Visual Summary

Scenario	Total Cost	New Facilities	Expansions	Status
Part 1 (Ideal)	\$430.1M	3,452	0	Feasible
Part 2 (Realistic)	\$444.9M	3,513	2,343	Feasible
Part 3 (Fairness)	Budget \$100M	varies	varies	Infeasible

4 Discussion

1. New Construction is More Cost-Effective

- Part 1 built **3,452 new** facilities, **0 expansions**.
- Part 2 built **3,513 new**, **2,343 expansions**.
- Baseline expansion cost and limits make expansion less flexible.
Policy Implication: Prioritize new construction to maximize coverage at scale.

2. Large Facilities Drive Efficiency

- **Large facilities** have lowest cost per slot ($\sim$ \$287 / child).
- Model naturally selects large facilities wherever possible.
Implication: Economies of scale dominate the solution.

3. Distance & Realism Increase Cost $\sim$ 3.5%

- Adding spatial and piecewise constraints raised cost from \$430M $\rightarrow$ \$445M.
- Real-world zoning and spatial fairness come at measurable fiscal cost.

4. Policy Constraint is Binding

- 1,323 ZIPs ($\approx$ 80%) exactly meet the 0–5 policy threshold (66.7%).
- This rule tightly controls resource allocation and drives much of the final cost.

5. Fairness Band Trade-off

- Strict $\delta = 0.10$ makes the model infeasible under \$100M.
- Slightly loosening δ (e.g., 0.12–0.15) *or* raising budget restores feasibility.
Policy Choice: Equity goals require fiscal flexibility.

Broader Reflection

This optimization demonstrates how **mathematical modeling can quantify policy trade-offs**:

- **Efficiency vs. Equity:** maximizing total coverage often conflicts with fairness constraints.
- Local realism (distance, zoning) drives up cost but improves accessibility.
- Expansion-only policies are ineffective at scale without budget flexibility.

Final message: Achieving universal childcare coverage is feasible but expensive. Policymakers must balance fairness goals, zoning realities, and fiscal limits.

Limitations

- Cost data are uniform — real costs vary by county and terrain.
- Facility sizes fixed (S, M, L) — no variable-size optimization.
- Population estimates assume 50% of 10–14 group are ages 10–12.
- Excludes ongoing operational expenses.
- Simplified fairness measure (coverage ratio only).

Next Steps

- Sensitivity tests: vary δ and budget.
- Add operating cost layer (multi-year budget optimization).
- Explore clustering or multi-objective weighting for urban vs. rural trade-offs.

Appendix

=====		
PART 1: IDEALISTIC SITUATION - RESULTS SUMMARY		
=====		
📊 COST BREAKDOWN:		

Expansion Costs:	\$	12,813,250.00
Construction Costs:	\$	358,540,000.00
Equipment Costs:	\$	58,786,200.00

TOTAL COST:	\$	430,139,450.00
=====		
🏢 FACILITY STATISTICS:		

New facilities built:		3,452
Facilities expanded:		0
New facilities by size:		
Size S:		658
Size M:		277
Size L:		2,517
📈 CAPACITY ADDITIONS:		

Expansion slots (0-12):		64,066
Expansion slots (0-5):		62,851
New facility slots (0-12):		1,128,000
New facility slots (0-5):		525,011
Total new capacity (0-12):		1,192,066
Total new capacity (0-5):		587,862
✅ COVERAGE STATISTICS:		

ZIPs meeting coverage threshold:		57
ZIPs meeting 0-5 policy threshold:		1,323
Total ZIPs covered:		1,646
=====		
✅ PART 1 OPTIMIZATION COMPLETE		
=====		

Figure 1: Results for Idealistic Scenario

COST BREAKDOWN:		
Expansion Costs:	\$	16,954,412.62
Construction Costs:	\$	369,205,000.00
Equipment Costs:	\$	58,786,200.00
TOTAL COST:	\$	444,945,612.62
FACILITY STATISTICS:		
Facilities expanded:		2,343
New facilities built:		3,513
New facilities by size:		
Size S (100 slots)		589
Size M (200 slots)		267
Size L (400 slots)		2,657
CAPACITY ADDITIONS:		
Expansion slots (0-12):		41,640
Expansion slots (0-5):		40,569
New facility slots (0-12):		1,175,100
New facility slots (0-5):		547,293
Total new capacity (0-12):		1,216,740
Total new capacity (0-5):		587,862
EXPANSION SEGMENT BREAKDOWN:		
Segment 1 (0-10%):		31,563 slots
Segment 2 (10-15%):		8,704 slots
Segment 3 (15-20%):		1,373 slots
Total expansion:		41,640 slots
COVERAGE STATISTICS:		
ZIPs meeting coverage threshold:		41
ZIPs meeting 0-5 policy threshold:		1,323
Total ZIPs covered:		1,646
DEMAND BREAKDOWN:		
High-demand ZIPs:		561
Normal-demand ZIPs:		1,085
PART 2 OPTIMIZATION COMPLETE		

Figure 2: Results for Realistic Scenario